

20251013EB_MN (vLH78)

Project: Yi-Shuan Tseng Holt Lab

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2025年10月13日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年10月15日 星期三

T2 -> T3

2025年10月17日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年10月20日 星期一

T4 -> T5

2025年10月21日 星期二

Prepare a plastic bottom 12-well plate to test Mazzoni protocol

1. dilute 300uL Matrigel into 30uL Neurobasal media
2. coat a full 12-well plastic bottom plate
3. incubate at 37C

---protocol reference-----

D8: Preparing surfaces for continued culture

Add 1:100 Matrigel (**not GFR**) diluted in Neurobasal medium to plates and incubate overnight at 37°C.

2025年10月22日 星期三

2/3: T5 -> T6

1/3: take 3mL of the EB to dissociate cells on day9 with Mazzoni lab protocol. (Mazzoni-lab_Generating motor neurons from human pluripotent stem cells (new))

--protocol and note-----

D9: MN progenitor stage and dissociation for subsequent culture

1. Swirl the plate or dish in a round motion to concentrate the floating embryoid bodies in the center of the wells.
2. Collect the EBs with a 5 or 10 mL pipette by aspirating from the center. (and transfer 3mL EB to a separate 15mL tube)
3. ~~Wash the plate or dish 1-2x with 2 mL DPBS and pool with previously collected medium.~~ (didn't do this)
4. Let EBs settle by gravity and aspirate supernatant.

5. Add 1 mL cold 1x Accutase and incubate for 10 min at RT.
6. Stop Accutase with 4 mL FBS serum.
7. Centrifuge 100g for 4 min. At this time, at the following to cold Neurobasal medium (make 4 mL per differentiation; if dissociating a large quantity of EBs, scale up).
 - a. 25 µg/mL DNase (For this step, I dissolve 1110u DNase I with 800uL Neurobasal media. Add 200uL into 5mL Neurobasal media to prepare a >50U/mL media. The calculation is according to the [STEMCELL website](#): 1mg/mL product. ≥ 2000 units per mg of protein)
8. Aspirate supernatant.
9. Add 500 µL cold Neurobasal medium with DNase.
10. Dissociate EBs with a 1 mL micropipette tip. Pipette up and down 8-10 times. Avoid bubbles.
11. Add 1 mL cold Neurobasal medium with DNase.
12. Let non-dissociated pieces of EBs settle to the bottom for 5-10 min.
13. Collect supernatant (1 mL) and transfer to 15 mL falcon tube.
14. Repeat steps 41-44 two more times (or more as needed). Pool supernatants of dissociated cells into 1 tube. (I did three rounds with 5min wait. I did the fourth pipetting and pool the media all together without wait)
15. Filter the dissociated cells through a 40 µm filter for cell culture into a new 15 mL falcon tube.
16. ~~Centrifuge the falcon tube at 300g for 4 min. Aspirate supernatant.~~
17. ~~Resuspend in Neurobasal to count cells.~~
18. ~~Count. Transfer desired number of cells to another 15 mL falcon tube (approximately 120k/cm² or fewer, depending on application).~~
 - a. ~~~7M in 100mm dish~~
 - b. ~~~2.5M in 60mm dish~~
19. Centrifuge cells at 300g for 4 min. At this time, add the following to N2B27 medium: (resuspend directly in 1mL T6. Pipette and add 2mL more T6.)
 - a. 500 µM AA (use 10uM AA as old protocol)
 - b. 100 nM RA
 - c. 500 nM SAG
 - d. 10 µM ROCK inhibitor Y-276329
 - e. 10 µM DAPT (leave out if OLIG2+/NKX2.2+ cells are desired)
20. Count cell

Table2

	A	B	C
1	conc. (K/mL)	survival	total (K in 3mL)
2	277	88	831

21. Aspirate Matrigel and plate.

Seed 1.5mL (831/2=415.5K) into 2 wells of 12well.

***I saw many clusters at the end. May need to pipette more rounds to get single cells. Also the cell count might not that accurate due to clusters. Emma did 10 rounds for her cells.

PlateA					^
	1	2	3	4	
A					
B					
C					

2025年10月24日 星期五

- ☒ 1/2: T6 -> T6 (~10mL for over-weekend)
- ☒ 1/2: Take 5mL (1/2) to dissociate into T7 with accutase
- ☒ Plate A: T7

2025年10月27日 星期一

T6 -> T7

Dissociation

- Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
- Remove the PLO and wash the plate with PBS once
- Coat the plate with laminine for 1hr.
- Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
- Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
- Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
- Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
- Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
- Filter the cell with 40um filter into a new 50mL tube
- Spin the cell down at 300rcf, 7min. Remove the supernatent.
- Add T7 (~5mL) to resuspend the cells and count it.
- Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P9	x5 (2/3)	

Table1			
	A	B	C
1	tube1	volume (mL)	temp
2	accutase	1	RT
3	HI-FBS/CTWM	2	
4	tube2		
5	0.25% trypsin	1	37C
6	HI-FBS/CTWM	2	

Table3										
	A	B	C	D	E	F	G	H	I	J
1	tube	conc. (K/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	1	146	21	1	146	A	24	211	4	6
3	2	698	99	1	698					
4					844					

Well1						
	1	2	3	4	5	6
A				vLH78 4uL in total		
B						
C						
D						

- 2025年10月28日 星期二
- Change T7 -> T7 if added virus
- 2025年10月29日 星期三
- T7-> T7 if not added virus
- 2025年10月30日 星期四
- T7 -> T8
- 2025年10月31日 星期五
- 1/2 T8 -> 1/2 T8

2025年11月3日 星期一

1/2 T8 -> 1/2 T8

2025年11月5日 星期三

1/2 T8 -> 1/2 T8

2025年11月6日 星期四

Treat cells at 4:00pm for the 24h timepoint

2025年11月7日 星期五

dilute 1:20 of SirDNA -> 0.32 in 6.4uL 0.15mM media --> 1.6 uL/well

Well2 ^

	1	2	3	4	5	6
A				1		
B				2		
C				3		
D				4		

microscope: new spinning disk confocal
100X, 1xzoom, 0.11um/px, 704x704
I tried to use 60x 1xzoom, the images were awful. It's hard to see GEMs. The visualization improved a lot with 100x oil objective!!!
USE 100X in the future!
NOTE: there were a lot of debris!!!!

Table4 ^

	A	B	C	D	E
1			78		
2			treatment	sirDNA	image
3	1	2.5mM 1h	13:45	13:45	14:30
4	2	0.15mM 1h	15:20	15:20	15:25
5	3	2.5mM 24h	11/6 16:00	13:45	15:50
6	4	0.15mM 24h	11/6 16:00	13:45	16:00

vLH78				
	A	B	C	D
1		laser power	exposure time	
2	488 10ms	100	10ms	movie
3	640	10	100ms	snapshot
4	488 search	50	50ms	snapshot



vLH78_fileid-conditions					
	A	B	C	D	E
1	image	treatment	treat time	type	virus
2	vLH78_MN_100x_1xzoom_0001	2.5mM gluc	1h	movie	78
3	vLH78_MN_100x_1xzoom_0002	2.5mM gluc	1h	snapshot	78
4	vLH78_MN_100x_1xzoom_0003	0.15mM gluc	1h	movie	78
5	vLH78_MN_100x_1xzoom_0004	0.15mM gluc	1h	snapshot	78
6	vLH78_MN_100x_1xzoom_0003_1	0.15mM gluc	1h	movie	78
7	vLH78_MN_100x_1xzoom_0004_1	0.15mM gluc	1h	snapshot	78
8	vLH78_MN_100x_1xzoom_0005	2.5mM gluc	24h	movie	78
9	vLH78_MN_100x_1xzoom_0006	2.5mM gluc	24h	snapshot	78
10	vLH78_MN_100x_1xzoom_0005_1	2.5mM gluc	24h	movie	78
11	vLH78_MN_100x_1xzoom_0006_1	2.5mM gluc	24h	snapshot	78
12	vLH78_MN_100x_1xzoom_0007	0.15mM gluc	24h	movie	78
13	vLH78_MN_100x_1xzoom_0008	0.15mM gluc	24h	snapshot	78



Mazzoni-lab_Generating motor neurons from human pluripotent stem cells (new)

Introduction

Efficient specification of spinal motor neuron subtypes from human pluripotent stem cells, adapted from Nedelec Lab (unpublished).

Materials

› Differentiation medium

- › 250 mL Advanced DMEM/F12
- › 250 mL Neurobasal medium
- › 5 mL Glutamax
- › 2 mL β -mercaptoethanol
- › 10 mL B27 supplement w/o Vitamin A
- › 5 mL N2 supplement

› Morphogens and inhibitors

- › Retinoic Acid (RA)
- › Smoothened Agonist (SAG)
- › ROCK inhibitor Y-27632 (prevent anoikis)
- › SB431542 (dual-SMAD inhibitor, drive toward neural progenitor)
- › LDN193189 (dual-SMAD inhibitor, drive toward neural progenitor)
- › CHIR (GSK3 inhibitor, activates Wnt, promotes differentiation)
- › DAPT (inhibits Notch signaling, drive toward neural phenotype)

› Neurotrophic factors

- › Ascorbic acid (AA)
- › BDNF
- › GDNF
- › CNTF

› Other reagents and materials

- › 1X PBS/DPBS
- › Accutase
- › DNase
- › Matrigel

- › FBS serum
- › 6-well ultra-low attachment dish
- › 40 µm filter for cell culture

Procedure

D0: Dissociation of hPSCs and EB formation while inducing differentiation

- ✓ 1. Aspirate hPSC culture medium.
- ✓ 2. Wash once with PBS.
- ✓ 3. Add cold Accutase to cover the cells for 3-4 min at 37°C until colonies detach.
Cells should be visible within the colonies but shouldn't be isolated floating cells.
- ✓ 4. Rinse the dish with E8 + rock inhibitor and collect floating colonies with a 5 mL pipette. Transfer into a 15 mL falcon tube.
- ✓ 5. Centrifuge cells at 200g for 4 min. Aspirate supernatant.
- ✓ 6. Add Neurobasal medium to resuspend the cells.
- ✓ 7. Count. Transfer desired number of cells to another 15 mL falcon tube (100-300k cells/mL depending on cell line, 2 mL per well of 6-well plate, 12 mL per 100mm dish).
For VChT-Cre tdTomato iPSC line, plate 200k/mL.
- ✓ 8. Centrifuge cells at 200g for 4 min. At this time, add the following to N2B27 medium:
 - 10 µM Rock inhibitor Y-27632
 - 20 µM SB431542
 - 0.1 µM LDN193189
 - 3 µM CHIR
- ✓ 9. Aspirate supernatant.
- ✓ 10. Add N2B27 medium with inhibitors to resuspend cells at the desired concentration.
- ✓ 11. Plate 2 mL per well in non-adherent 6-well dish or 12 mL in 100mm suspension culture dish.

D2: Continuing differentiation with addition of morphogens

- ✓ 12. Swirl the plate or dish in a round motion to concentrate the floating embryoid bodies in the center of the wells.

- ✓ 13. Collect the EBs with a 5 or 10 mL pipette by aspirating from the center.
- ✓ 14. Transfer the supernatant with floating EBs to a 15 mL falcon tube.
- ✓ 15. Wash the plate or dish 1-2x with 2 mL DPBS and pool with previously collected medium.
- ✓ 16. Centrifuge 100g for 4 min. At this time, add the following to N2B27 medium:
 - 20 μ M SB431542
 - 0.1 μ M LDN193189
 - 3 μ M CHIR
 - 100 nM RA
 - 500 nM SAG
- ✓ 17. Carefully aspirate the supernatant.
- ✓ 18. Resuspend EBs in N2B27 medium with morphogens and plate.

D4: Differentiation toward MN progenitor fate

- ✓ 19. Swirl the plate or dish in a round motion to concentrate the floating embryoid bodies in the center of the wells.
- ✓ 20. Collect the EBs with a 5 or 10 mL pipette by aspirating from the center.
- ✓ 21. Wash the plate or dish 1-2x with 2 mL DPBS and pool with previously collected medium.
- ✓ 22. Centrifuge 100g for 4 min. At this time, add the following to N2B27 medium:
 - 20 μ M SB431542
 - 0.1 μ M LDN193189
 - 100 nM RA
 - 500 nM SAG
- ✓ 23. Carefully aspirate the supernatant.
- ✓ 24. Resuspend EBs in N2B27 medium with morphogens and plate.

D7: Differentiation toward MN progenitor fate

- ✓ 25. Swirl the plate or dish in a round motion to concentrate the floating embryoid bodies in the center of the wells.
- ✓ 26. Collect the EBs with a 5 or 10 mL pipette by aspirating from the center.
- ✓ 27. Wash the plate or dish 1-2x with 2 mL DPBS and pool with previously collected medium.

- ✓ 28. Let EBs settle by gravity or centrifuge 100g for 4 min. At this time, add the following to N2B27 medium:
 - 100 nM RA
 - 500 nM SAG
- ✓ 29. Carefully aspirate the supernatant.
- ✓ 30. Resuspend EBs in N2B27 medium with morphogens and plate.

D8: Preparing surfaces for continued culture

- ✓ 31. Add 1:100 Matrigel (**not GFR**) diluted in Neurobasal medium to plates and incubate overnight at 37°C.

D9: MN progenitor stage and dissociation for subsequent culture

- ✓ 32. Swirl the plate or dish in a round motion to concentrate the floating embryoid bodies in the center of the wells.
- ✓ 33. Collect the EBs with a 5 or 10 mL pipette by aspirating from the center.
- ✓ 34. Wash the plate or dish 1-2x with 2 mL DPBS and pool with previously collected medium.
- ✓ 35. Let EBs settle by gravity and aspirate supernatant.
- ✓ 36. Add 1 mL cold 1x Accutase and incubate for 10 min at RT.
- ✓ 37. Stop Accutase with 4 mL FBS serum.
- ✓ 38. Centrifuge 100g for 4 min. At this time, add the following to cold Neurobasal medium (make 4 mL per differentiation; if dissociating a large quantity of EBs, scale up).
 - 25 µg/mL DNase
- ✓ 39. Aspirate supernatant.
- ✓ 40. Add 500 µL cold Neurobasal medium with DNase.
- ✓ 41. Dissociate EBs with a 1 mL micropipette tip. Pipette up and down 8-10 times. Avoid bubbles.
- ✓ 42. Add 1 mL cold Neurobasal medium with DNase.
- ✓ 43. Let non-dissociated pieces of EBs settle to the bottom for 5-10 min.
- ✓ 44. Collect supernatant (1 mL) and transfer to 15 mL falcon tube.
- ✓ 45. Repeat steps 41-44 two more times (or more as needed). Pool supernatants of dissociated cells into 1 tube.

- ✓ 46. Filter the dissociated cells through a 40 µm filter for cell culture into a new 15 mL falcon tube.
- ✓ 47. Centrifuge the falcon tube at 300g for 4 min. Aspirate supernatant.
- ✓ 48. Resuspend in Neurobasal to count cells.
- ✓ 49. Count. Transfer desired number of cells to another 15 mL falcon tube (approximately 120k/cm² or fewer, depending on application).
 - ~7M in 100mm dish
 - ~2.5M in 60mm dish
- ✓ 50. Centrifuge cells at 300g for 4 min. At this time, add the following to N2B27 medium:
 - 500 µM AA
 - 100 nM RA
 - 500 nM SAG
 - 10 µM ROCK inhibitor Y-276329
 - 10 µM DAPT (leave out if OLIG2+/NKX2.2+ cells are desired)
- ✓ 51. Aspirate Matrigel and plate.

D11: Differentiation toward post-mitotic MN

- ✓ 52. Add the following to N2B27 medium:
 - 500 µM AA
 - 100 nM RA
 - 500 nM SAG
 - 10 µM DAPT (leave out if OLIG2+/NKX2.2+ cells are desired)
 - 10 ng/mL BDNF
 - 10 ng/mL GDNF
 - 10 ng/mL CNTF
- ✓ 53. Change media with N2B27 with neurotrophic factors added.

D14: Post-mitotic motor neurons

- ✓ 54. Add the following to N2B27 medium:
 - 500 µM AA
 - 10 ng/mL BDNF
 - 10 ng/mL GDNF
 - 10 ng/mL CNTF
- ✓ 55. Change media with N2B27 with neurotrophic factors added.